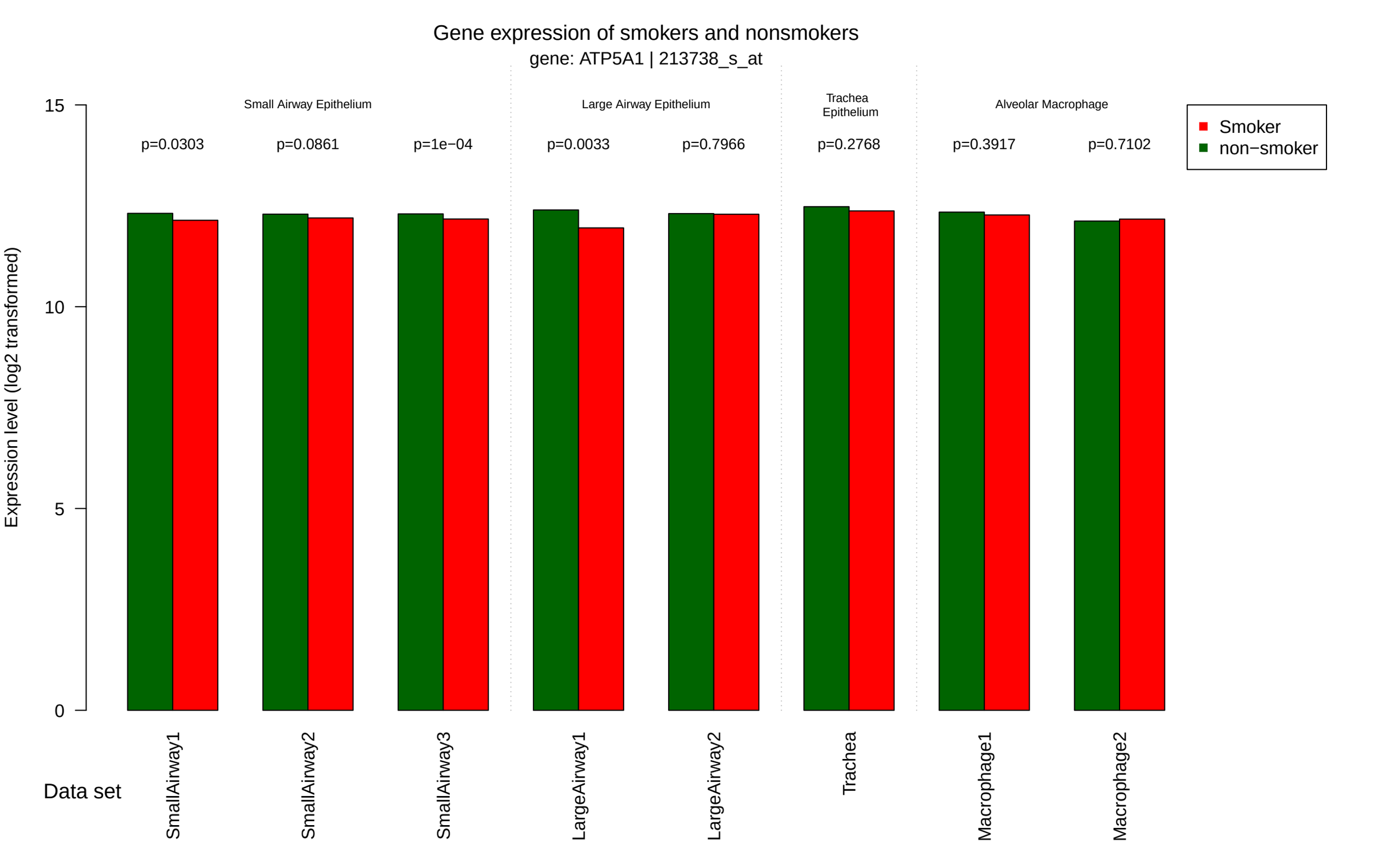
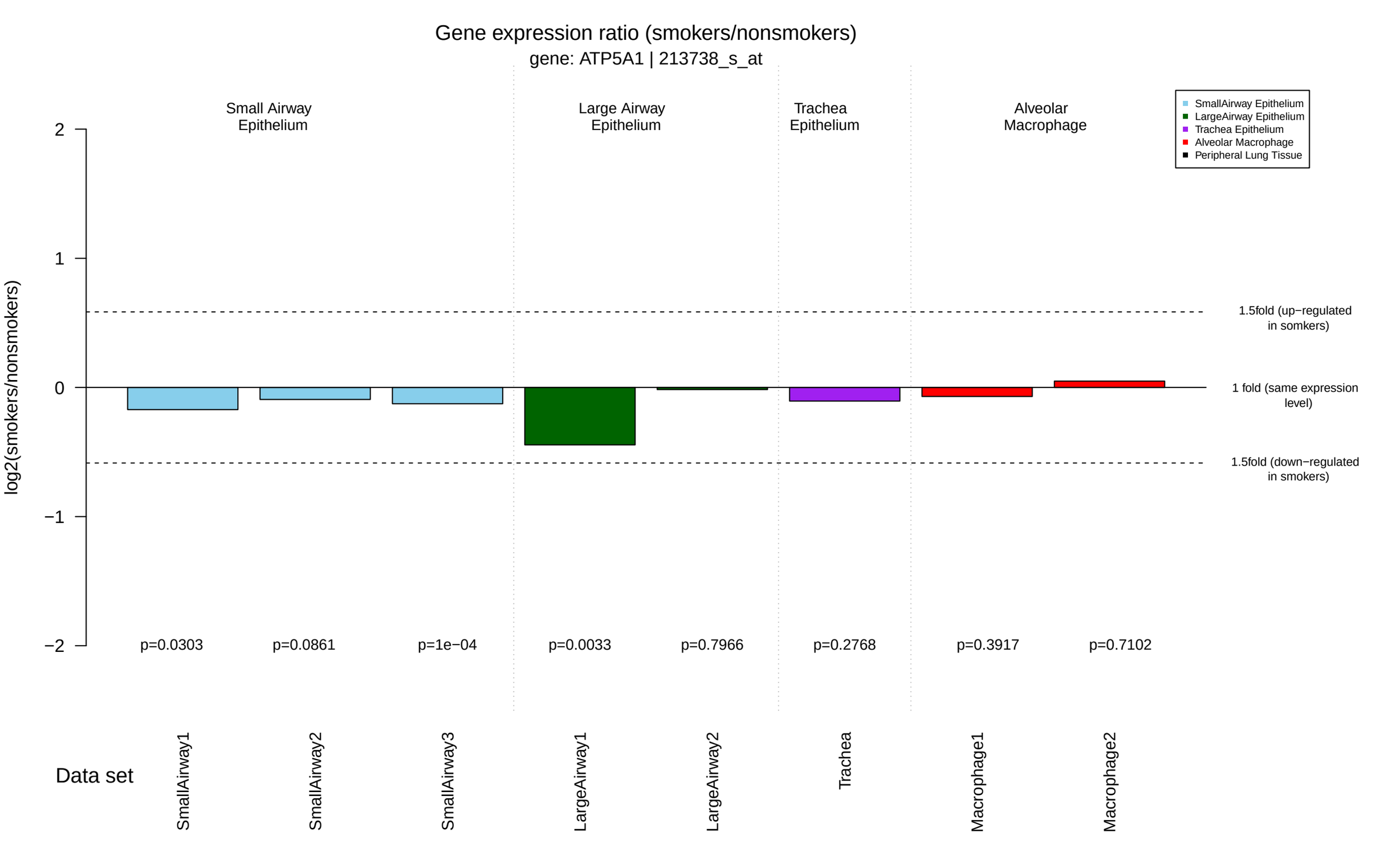
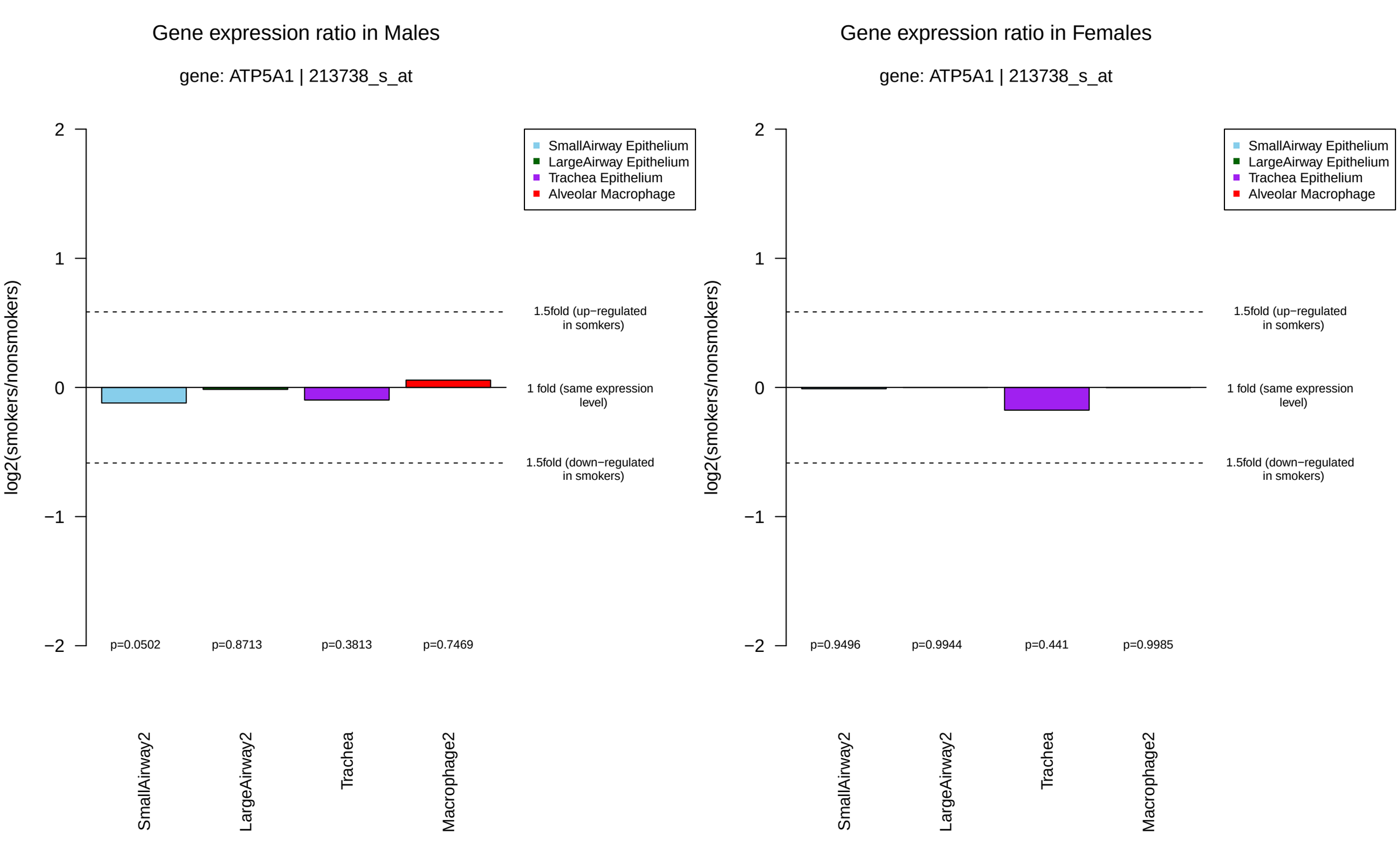
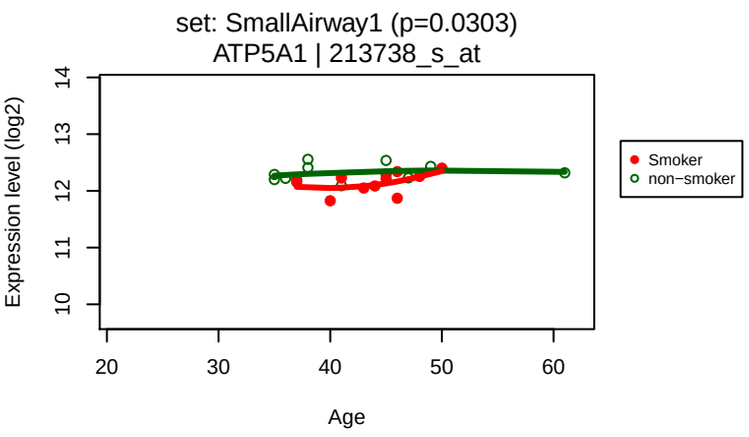
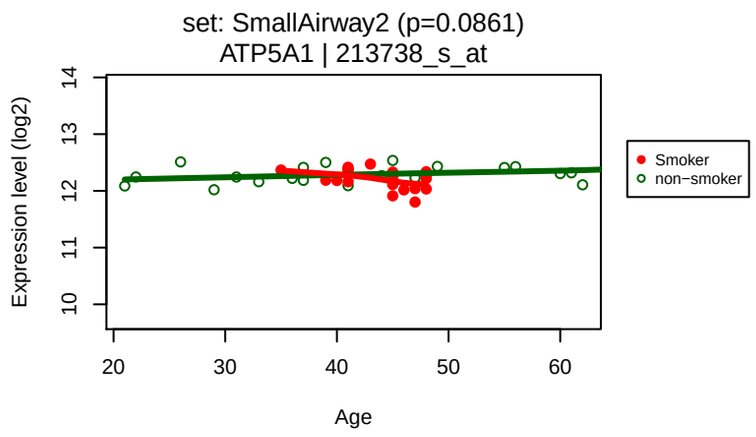
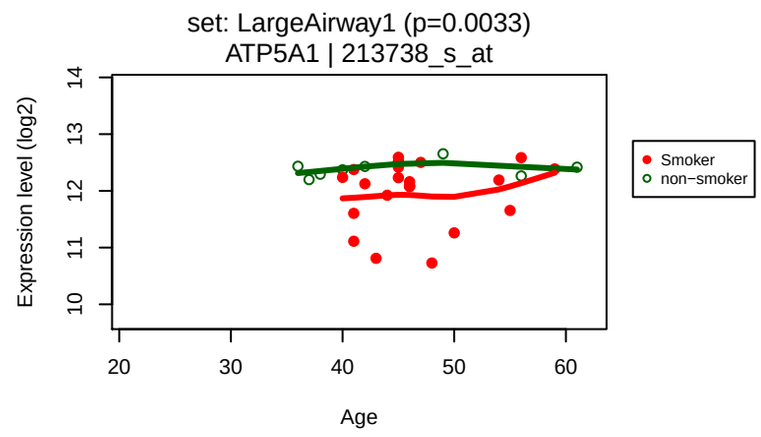

Gene expression of smokers and nonsmokers
gene: ATP5A1 | 213738_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0303
p=0.0861
p=1e−04
p=0.0033
p=0.7966
p=0.2768
p=0.3917
p=0.7102
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

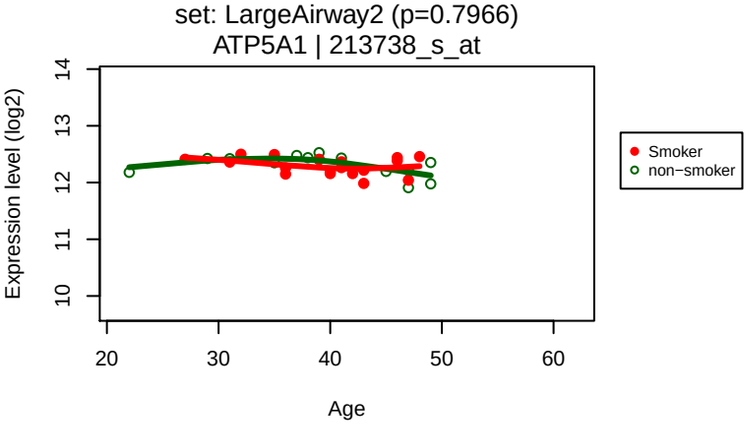

Gene expression ratio (smokers/nonsmokers)
gene: ATP5A1 | 213738_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0303
p=0.0861
p=1e−04
p=0.0033
p=0.7966
p=0.2768
p=0.3917
p=0.7102
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

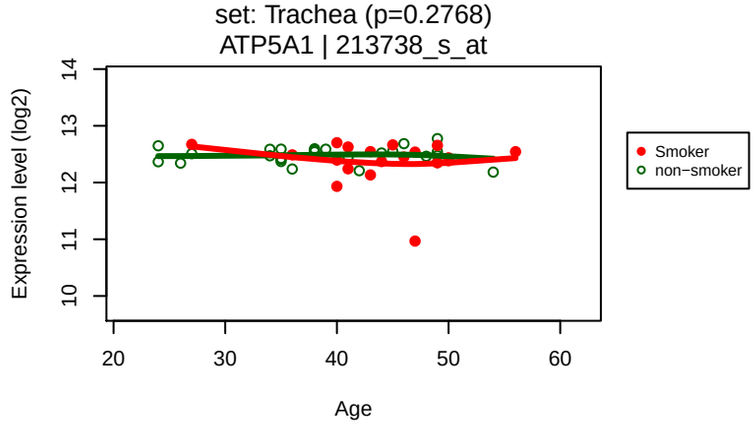

Gene expression ratio in Males
gene: ATP5A1 | 213738_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0502
p=0.8713
p=0.3813
p=0.7469
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP5A1 | 213738_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9496
p=0.9944
p=0.441
p=0.9985
SmallAirway2
LargeAirway2
Trachea
Macrophage2

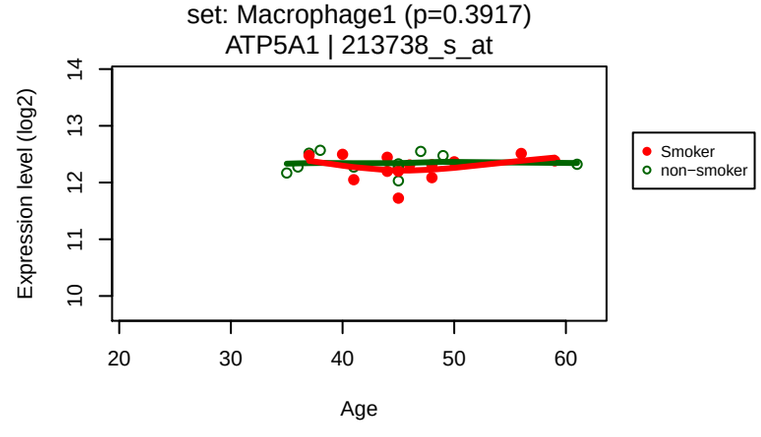

set: SmallAirway1 (p=0.0303)
ATP5A1 | 213738_s_at
Expression level (log2)
Age
Smoker
non−smoker

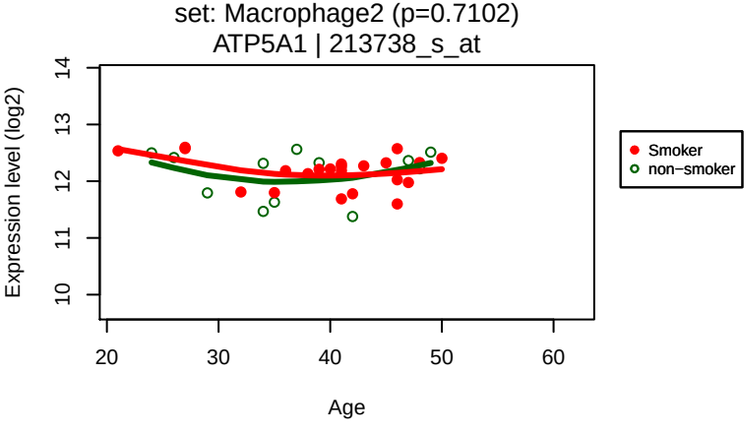

set: SmallAirway2 (p=0.0861)
ATP5A1 | 213738_s_at
Expression level (log2)
Age
Smoker
non−smoker

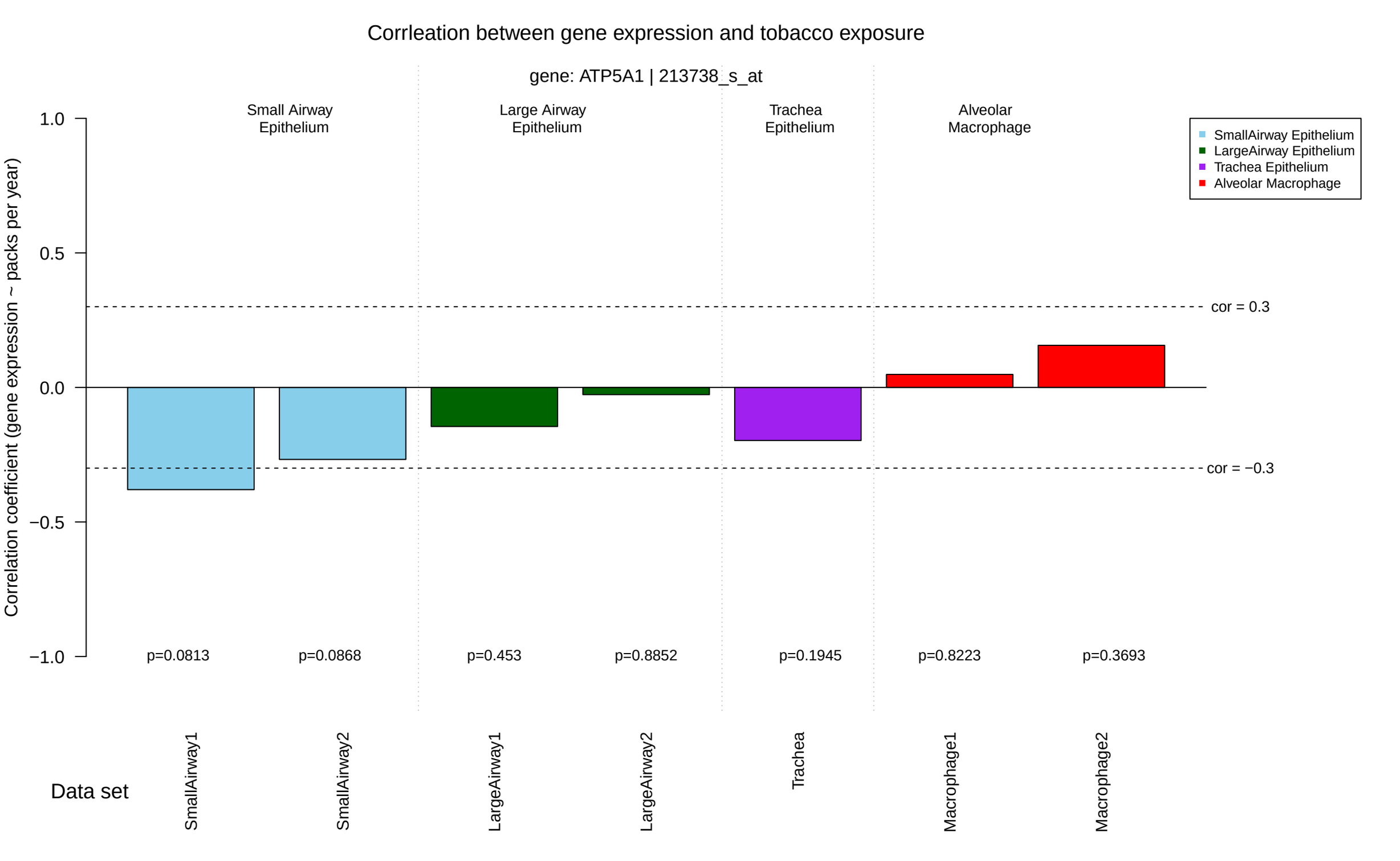

set: LargeAirway1 (p=0.0033)
ATP5A1 | 213738_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7966)
ATP5A1 | 213738_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2768)
ATP5A1 | 213738_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3917)
ATP5A1 | 213738_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7102)
ATP5A1 | 213738_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP5A1 | 213738_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0813
p=0.0868
p=0.453
p=0.8852
p=0.1945
p=0.8223
p=0.3693
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2